AI-assisted multi-agent quantitative fund

FE5209 project scope and design

Three strategy PMs managing a USD 5m paper fund

Proposed design
13 October 2026

Project scope

A portfolio experiment with a visible AI review workflow

Course requirements

- Automated model updates and daily forecasts
- Mandatory class assets with eligible history
- Out-of-sample evaluation: 1–23 October 2026
- Code, closing-price CSV and presentation

Proposed first version

- Three PMs with separate paper ledgers
- Shared allocation and fund risk controls
- AI agents inspect tools and cite evidence
- One hosted application with saved run history

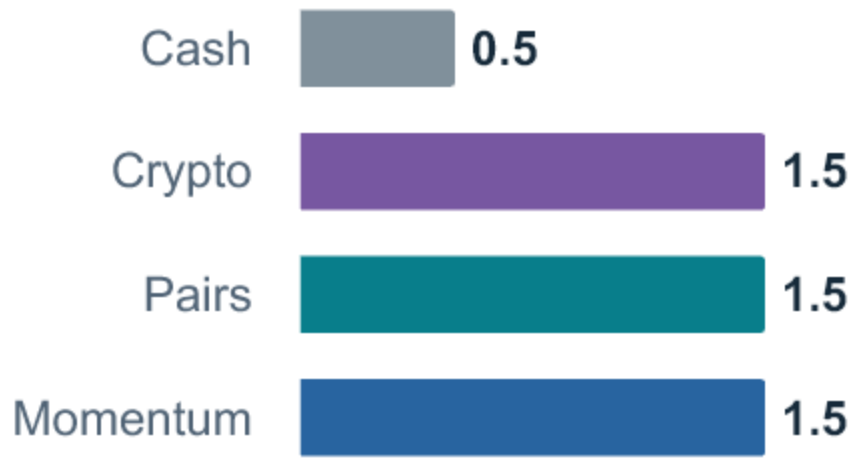
Initial capital and execution assumptions need instructor confirmation

USD 5m fund allocation

Illustrative initial mandates, with USD 0.5m held centrally

Capital owner	USD	Share
Momentum PM	1,500,000	30%
Pairs PM	1,500,000	30%
Crypto PM	1,500,000	30%
Central cash	500,000	10%
Total fund	5,000,000	100%

Initial capital, USD millions



The baseline keeps capital in each sleeve. Returns cause fund weights to drift.

Equal initial capital can produce different risk. Each PM can retain cash.

The multi-AI contribution

Each PM combines its existing strategy with a planned tool-using AI representative

AI representative	Diagnostic tools	Evidence-linked output
Momentum PM AI	Read rankings, holdings and sector exposure	Explain selections and concentration concerns
Pairs PM AI	Read daily spread signals, hedge ratios and pair health	Explain both legs and relationship concerns
Crypto PM AI	Read signals, volatility, liquidity and trading costs	Explain exposure or cash and data concerns

AI chooses investigations and writes PM memos

Quantitative code calculates targets. The coordinator and risk controller govern execution.

Daily pairs decision cycle

Pairs frequency is confirmed. Exact signal and fill timing remains to be agreed.

When	Owner	Output
Session t, after close	Data adapter	Completed prices and validated timestamps
Session t, after close	Pairs quant engine	Daily model update and linked target legs
Before execution	Coordinator and risk	Combined exposure and approval verdict
Next eligible session	Paper executor	Simulated fills under the declared cost model
After valuation	Accounting and AI PMs	Fund NAV, CSV and evidence-linked memos

Both legs retain their hedge relationship. A daily signal can lead to no trade.

Capital, risk and accounting

The coordinator proposes budgets. The risk controller has final approval.

Portfolio coordinator

- Assign the initial PM capital mandates
- Combine holdings and identify overlap
- Track PM contribution to fund P&L
- Log transfers separately from returns

Independent risk controller

- Check concentration and gross/net exposure
- Preserve linked pair positions
- Check available cash, margin and data
- Reject breaches and record the reason

Fund NAV = sum of PM NAVs + central cash

Capital transfers preserve fund NAV. Costs enter the ledger once.

Evaluation plan

Trading performance and AI review quality have separate tests

Experiment	Question	Main evidence
Each PM alone	Profitable after costs?	Net return, Sharpe and drawdown
Initial-budget blend	Better as a combined fund?	Fund NAV, correlation and PM P&L
Allocation alternative	Better after reallocation costs?	Incremental return, risk and turnover
AI enabled / disabled	Accurate and useful explanations?	Cited facts, review time and cost

The AI comparison keeps quantitative portfolio decisions identical

Use the official benchmark and consistent dates, costs and capital conventions.

Delivery and ownership

One application hosts the PM views and the daily decision history

Submission and demonstration

- Reproducible strategy code and closing-price CSV
- Fund and PM performance dashboard
- Actual tool calls, risk verdicts and PM memos
- An isolated replay of a selected historical day

Proposed hosting

- Streamlit dashboard
- One Python engine with a scheduled runner
- SQLite on persistent storage
- Server-side data and model credentials

Ownership: three PM owners, an integration lead and a reporting/operations lead

For four members, share reporting and operations. Final slides must name every member.

Readiness and remaining inputs

The combined fund and connected AI PMs are proposed work

Existing foundation

- Crypto data and daily paper workflow
- Agent orchestration, risk checks and logs
- Recorded intents and optional AI replies

Inputs before integration

- Strategy packages, assets and momentum cadence
- Mandatory assets and official benchmark
- Capital, costs, shorting and execution rules
- Member names and individual responsibilities

First version: initial budgets, shared accounting, fund risk and AI tool access

Extensions: dynamic allocation, regime routing and physical order netting